

Term Project Proposal

Energy-Robustness Trade-offs in Compressed TinyML Vision Models

CSE407 [Green Computing]

Section: 3 [Summer 2026]

Group: 07

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Problem Statement

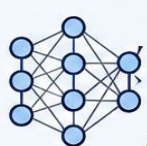
Problem

We do not know whether compressing a CNN to reduce energy consumption also reduces its robustness under real-world corrupted conditions, because existing studies mainly evaluate compression ratio rather than actual measured energy.

Core Research Problem

For similar energy reductions across different compression techniques, which method best preserves robustness while maintaining acceptable accuracy?

1. Baseline Model



f
ResNet-18
(CIFAR-10)

Clean Acc: $Acc(f)$



2. Compress



Quantize
or
Prune
 (c_i, α)



3. Measure



Energy $E(c_i, \alpha)$



Accuracy $Acc(c_i, \alpha)$



Robustness $mCE(c_i, \alpha)$

Motivation

Why it matters

- ICT contributes **2–3% of global CO₂ emissions** (projected **7–8% by 2030**).
- AI inference is becoming a major source of carbon emissions.
- Edge/TinyML devices require **energy-efficient model compression**.
- Real-world inputs contain **noise, blur, fog, and compression artifacts**.
- **Risk:** Efficient models may lose reliability in real deployment.

Research Gap

- Existing work compares **compression vs. robustness** using **compression ratio**, not real energy.
- Most studies evaluate **only one compression technique** (e.g., pruning or quantization).
- Real hardware energy studies **do not evaluate robustness**.



Why This Study

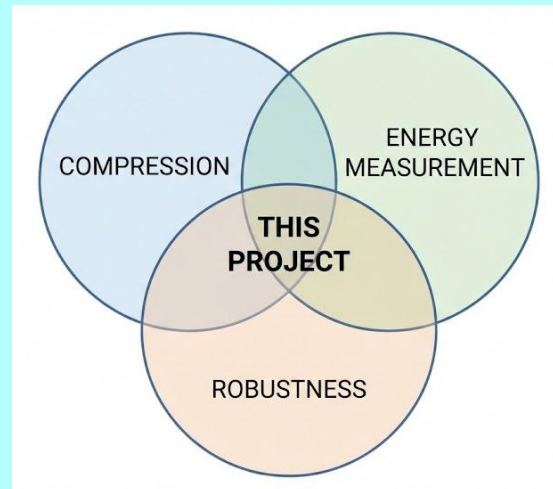
- Compression ratio $\neq$ actual energy consumption.
- Energy depends on **hardware and runtime behavior**.
- Requires **direct hardware-based energy measurement** with robustness evaluation.

Literature Gap & Why existing work is insufficient

Key Studies

- **Da Silva et al. (2025):** Compression vs. robustness (*Base Paper*)
- **Xiao et al. (2023):** Quantization & robustness
- **Mitra et al. (2024):** Pruning & robustness
- **Husom et al. (2025):** Hardware energy measurement (LLMs)
- **Hendrycks & Dietterich (2019):** CIFAR-10-C & mCE benchmark

Gap: no major existing study combines measured energy, corruption robustness, and multiple compression techniques for vision models



Existing Research

Three research areas exist separately:

- **Compression vs. robustness**
- **Hardware-based energy measurement**
- **Compression benchmarking**

Research Questions

Primary RQ:

For TinyML-scale CNNs, how do different compression techniques, such as quantization, magnitude pruning, and structured pruning, that achieve comparable energy reductions differ in their impact on robustness to natural image corruption?

Sub-Research Questions:

1. At comparable energy reductions, does robustness (mCE) differ significantly across the three compression techniques?
2. Which technique gives the best overall balance of energy savings, accuracy, and robustness?

Base Paper and Related Work

Study	Techniques Compared	Robustness Type	Measured Energy?	TinyML-scale Focus?
Da Silva et al. [1] (Base Paper)	Quant. + Pruning + Clustering	Natural corruption	No (ratio only)	Partial (MobileNetV2)
Xiao et al. [2]	Quantization only	Natural + adversarial	No	No
Mitra et al. [3]	Pruning only	Natural corruption + calibration	No	No
Husom et al. [4]	Quantization (LLMs)	Not studied	Yes (hardware)	Edge, not TinyML-CNN
This Project	Quant. + 2 Pruning types	Natural corruption	Yes (hardware-measured)	Yes (TinyML-scale CNN)

Methodology, Setup

Baseline Model

- **ResNet-18** trained on **CIFAR-10**

Compression Techniques

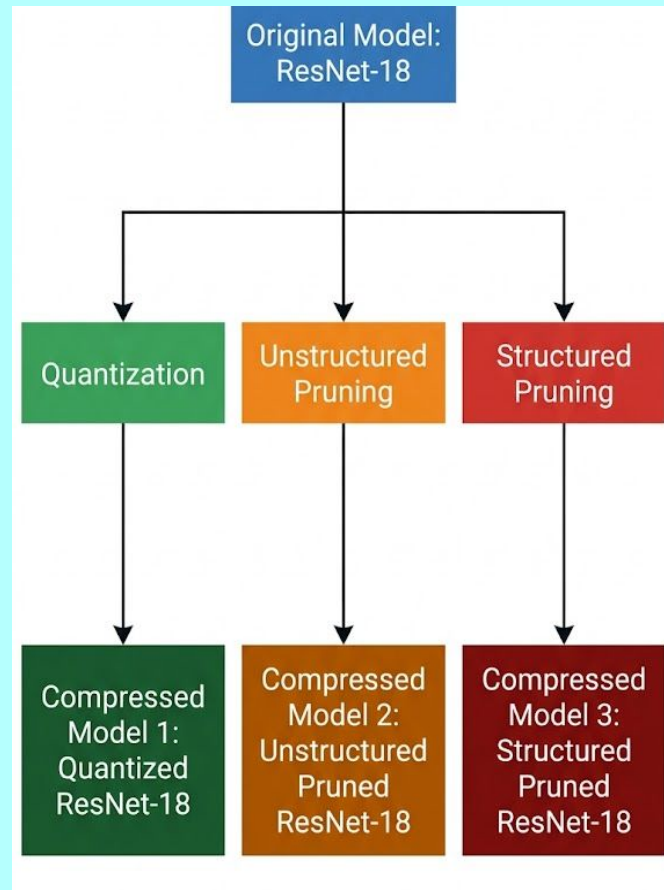
- **T1:** INT8 post-training quantization
- **T2:** Unstructured magnitude pruning (30%, 50%, 70% sparsity)
- **T3:** Structured channel pruning (30%, 50%, 70% sparsity)

Framework

- **PyTorch** (recommended by the base paper for future work)

Training

- **Adam** optimizer
- **Cross-entropy** loss
- **Early stopping** (patience = 5)
- **Maximum 30 epochs**



Methodology (Continued)

Energy Measurement

- **Hardware:** Kaggle GPU
- **Tools:** pynvml, nvidia-smi, validated with CodeCarbon
- **Protocol:** 10 warm-up + 30 inference runs
- **Metric:** Joules/inference & % energy reduction

Robustness Evaluation

- **Dataset:** CIFAR-10-C
- **Corruptions:** Noise, blur, fog, JPEG artifacts
- **Metric:** mCE (Mean Corruption Error)

Fair Comparison

- Compare all techniques at **matched energy reduction levels**



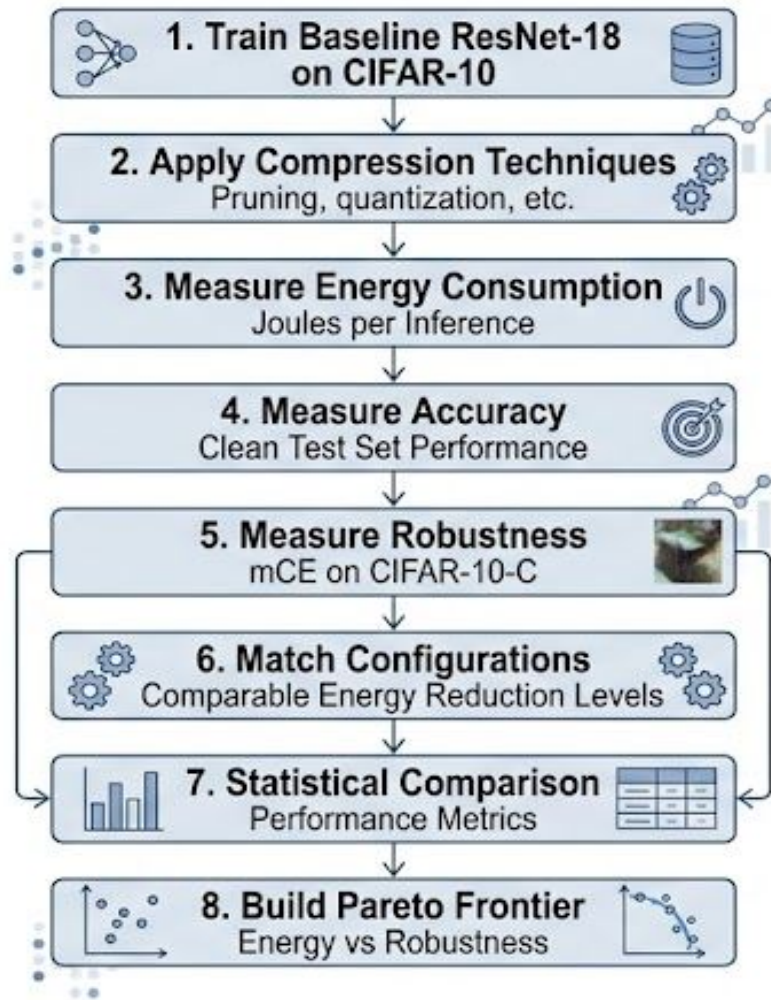
Statistical Analysis

- **Shapiro–Wilk & Levene's tests**
- **ANOVA + Tukey** (if assumptions hold)
- **Kruskal–Wallis + Dunn's** (otherwise)

Methodology Pipeline and Expected Outcome

Expected Outcomes

- Energy-robustness Pareto frontier for compression techniques.
- Best compression method for TinyML and edge deployment.
- Determine whether compression ratio is a reliable proxy for real energy, or if direct energy measurement is required.



References and Links

1. Da Silva, Pereira, Barboza, dos Santos Neto, Ribeiro (2025), Evaluating the Impact of Compression Techniques on the Robustness of CNNs under Natural Corruptions. arxiv.org/abs/2512.24971
2. Xiao, Liu, Zhang, Qin, Guo, Liu (2023), RobustMQ, Benchmarking Robustness of Quantized Models, Visual Intelligence. arxiv.org/abs/2308.02350
3. Mitra, Schwalbe, Klein (2024), Investigating Calibration and Corruption Robustness of Post-hoc Pruned Perception CNNs, CVPR Workshops. arxiv.org/abs/2405.20876
4. Husom, Goknil, Astekin, Shar, Kasen, Sen, Mithassel, Soylu (2025), Sustainable LLM Inference for Edge AI, ACM Transactions on Internet of Things. arxiv.org/abs/2504.03360
5. Hendrycks, Dietterich (2019), Benchmarking Neural Network Robustness to Common Corruptions and Perturbations, ICLR. arxiv.org/abs/1903.12261